Q1 - 2024 (01 Feb Shift 1)

If the shortest distance between the lines $\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 1 , then the sum of all possible values of λ is :

(1) 0

(2) $2\sqrt{3}$

(3) $3\sqrt{3}$

(4) $-2\sqrt{3}$

Q2 - 2024 (01 Feb Shift 1)

Let the line of the shortest distance between the lines

$L_1 : \vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and

$\mathrm{L}_2 : \overrightarrow{\mathrm{r}}=(4\hat{\mathrm{i}}+5\hat{\mathrm{j}}+6\hat{\mathrm{k}})+\mu(\hat{\mathrm{i}}+\hat{\mathrm{j}}-\hat{\mathrm{k}})$

intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the midpoint of the line segment PQ, then $2(\alpha+\beta+\gamma)$ is equal to

Q3 - 2024 (01 Feb Shift 2)

Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point $R(1,2,3)$. If the centroid of the triangle PQR is (α, β, γ), then $\alpha^2+\beta^2+\gamma^2$ is:

(1) 26

(2) 36

(3) 18

(4) 24

Q4 - 2024 (01 Feb Shift 2)

If the mirror image of the point $\mathrm{P}(3,4,9)$ in the line $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ is (α, β, γ), then $14(\alpha+\beta+\gamma)$ is :

(1) 102

(2) 138

(3) 108

(4) 132

Q5 - 2024 (27 Jan Shift 1)

The distance, of the point $(7,-2,11)$ from the line $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$ along the line $\frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$, is :

(1) 12

(2) 14

(3) 18

(4) 21

Q6 - 2024 (27 Jan Shift 1)

If the shortest distance between the lines $\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$ and $\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

(1) 5

(2) 8

(3) 7

(4) 10

Q7 - 2024 (27 Jan Shift 2)

Let the image of the point $(1,0,7)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α,β,γ). Then which one of the following points lies on the line passing through (α,β,γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

(1) $(1,-2,1+\sqrt{2})$

(2) $(1,2,1-\sqrt{2})$

(3) $(3,4,3-2\sqrt{2})$

(4) $(3,-4,3+2\sqrt{2})$

Q8 - 2024 (27 Jan Shift 2)

The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let l denotes the length of the angle bisector AD of $\angle$BAC where D is on the line segment BC, then $2l^2$ equals :

(1) 49

(2) 42

(3) 50

(4) 45

Q9 - 2024 (27 Jan Shift 2)

Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $l_1^2+l_2^2+l_3^2$ equals:

(1) $\frac{1}{5}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

Q10 - 2024 (27 Jan Shift 2)

The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to

Q11 - 2024 (29 Jan Shift 1)

Let O be the origin and the position vector of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of $\angle AOB$ meets the line AB at C, then the length of OC is

(1) $\frac{2}{3}\sqrt{31}$

(2) $\frac{2}{3}\sqrt{34}$

(3) $\frac{3}{4}\sqrt{34}$

(4) $\frac{3}{2}\sqrt{31}$

Q12 - 2024 (29 Jan Shift 1)

Let PQR be a triangle with $R(-1,4,2)$. Suppose $M(2,1,2)$ is the mid point of PQ. The distance of the centroid of $\triangle$PQR from the point of intersection of the line $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ and $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ is

(1) 69

(2) 9

(3) $\sqrt{69}$

(4) $\sqrt{99}$

Q13 - 2024 (29 Jan Shift 1)

A line with direction ratios $2,1,2$ meets the lines $x=y+2=z$ and $x+2=2y=2z$ respectively at the point P and Q. if the length of the perpendicular from the point $(1,2,12)$ to the line PQ is l, then l^2 is

Q14 - 2024 (29 Jan Shift 2)

Let $\mathrm{P}(3,2,3)$, $\mathrm{Q}(4,6,2)$ and $\mathrm{R}(7,3,2)$ be the vertices of $\triangle$PQR. Then, the angle $\angle$QPR is

(1) $\frac{\pi}{6}$

(2) $\cos^{-1}\left(\frac{7}{18}\right)$

(3) $\cos^{-1}\left(\frac{1}{18}\right)$

(4) $\frac{\pi}{3}$

Q15 - 2024 (29 Jan Shift 2)

Let O be the origin, and M and N be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{\mathrm{x}+8}{12}=\frac{\mathrm{y}+2}{5}=\frac{\mathrm{z}+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{\mathrm{OM}}\cdot\overrightarrow{\mathrm{ON}}$ is equal to ______.

Q16 - 2024 (30 Jan Shift 1)

Let (α, β, γ) be the foot of perpendicular from the point $(1,2,3)$ on the line $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. then $19(\alpha+\beta+\gamma)$ is equal to :

(1) 102

(2) 101

(3) 99

(4) 100

Q17 - 2024 (30 Jan Shift 1)

If d_1 is the shortest distance between the lines $x+1=2y=-12z, x=y+2=6z-6$ and d_2 is the shortest distance between the lines $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}, \frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}\,\mathrm{d}_1}{\mathrm{d}_2}$ is :

Q18 - 2024 (30 Jan Shift 2)

Let $L_1 : \overrightarrow{\mathbf{r}}=(\hat{i}-\hat{j}+2\hat{k})+\lambda(\hat{i}-\hat{j}+2\hat{k}), \lambda \in R$ $L_2 : \overrightarrow{\mathbf{r}}=(\hat{\mathbf{j}}-\hat{\mathbf{k}})+\mu(3\hat{\mathbf{i}}+\hat{\mathbf{j}}+\mathrm{p}\hat{\mathbf{k}}), \mu \in R$ and $\mathrm{L}_3 : \overrightarrow{\mathbf{r}}=\delta(\ell\hat{\mathbf{i}}+\mathrm{m}\hat{\mathbf{j}}+\mathrm{n}\hat{\mathbf{k}}) \delta \in \mathrm{R}$

Be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2. Then the point which lies on L_3 is

(1) $(-1,7,4)$

(2) $(-1,-7,4)$

(3) $(1,7,-4)$

(4) $(1,-7,4)$

Q19 - 2024 (30 Jan Shift 2)

Let a line passing through the point $(-1,2,3)$ intersect the lines $L_1 : \frac{x-1}{3}=\frac{y-2}{2}=\frac{z+1}{-2}$ at $M(\alpha, \beta, \gamma)$ and $L_2 : \frac{x+2}{-3}=\frac{y-2}{-2}=\frac{z-1}{4}$ at $N(a, b, \mathrm{c})$. Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equals______.

Q20 - 2024 (31 Jan Shift 1)

The distance of the point $Q(0,2,-2)$ form the line passing through the point $\mathrm{P}(5,-4,3)$ and perpendicular to the lines $\overrightarrow{\mathrm{r}}=(-3\hat{\mathrm{i}}+2\hat{\mathrm{k}})+\lambda(2\hat{i}+3\hat{j}+5\hat{k}), \quad \lambda\in\mathbb{R}$ and $\vec{r}=(\hat{i}-2\hat{j}+\hat{k})+$ $\mu(-\hat{\mathbf{i}}+3\hat{\mathbf{j}}+2\hat{\mathbf{k}}), \mu\in\mathbb{R}$ is

(1) $\sqrt{86}$

(2) $\sqrt{20}$

(3) $\sqrt{54}$

(4) $\sqrt{74}$

Q21 - 2024 (31 Jan Shift 1)

Let Q and R be the feet of perpendiculars from the point $P(a,a,a)$ on the lines $x=y, z=1$ and $x=-y$, $z=-1$ respectively. If $\angle QPR$ is a right angle, then $12\mathrm{a}^2$ is equal to _______.

Q22 - 2024 (31 Jan Shift 2)

Let (α,β,γ) be mirror image of the point $(2,3,5)$ in the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$.

Then $2\alpha+3\beta+4\gamma$ is equal to

(1) 32

(2) 33

(3) 31

(4) 34

Q23 - 2024 (31 Jan Shift 2)

The shortest distance between lines L_1 and L_2, where $L_1:\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ and L_2 is the line passing through the points $A(-4,4,3)\cdot B(-1,6,3)$ and perpendicular to the line $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{1}$, is

(1) $\frac{121}{\sqrt{221}}$

(2) $\frac{24}{\sqrt{117}}$

(3) $\frac{141}{\sqrt{221}}$

(4) $\frac{42}{\sqrt{117}}$

Q24 - 2024 (31 Jan Shift 2)

A line passes through $A(4,-6,-2)$ and $B(16,-2,4)$. The point $\mathrm{P}(\mathrm{a}, \mathrm{b}, \mathrm{c})$ where $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points $P(a, b, c)$ and $Q(4,-12,3)$ is equal to _______.

Answer Key

Q1 (2)	**Q2** (21)	**Q3** (3)	**Q4** (3)
Q5 (2)	**Q6** (2)	**Q7** (3)	**Q8** (4)
Q9 (2)	**Q10** (108)	**Q11** (2)	**Q12** (3)
Q13 (65)	**Q14** (4)	**Q15** (9)	**Q16** (2)
Q17 (16)	**Q18** (1)	**Q19** (196)	**Q20** (4)
Q21 (12)	**Q22** (2)	**Q23** (3)	**Q24** (22)

Q1

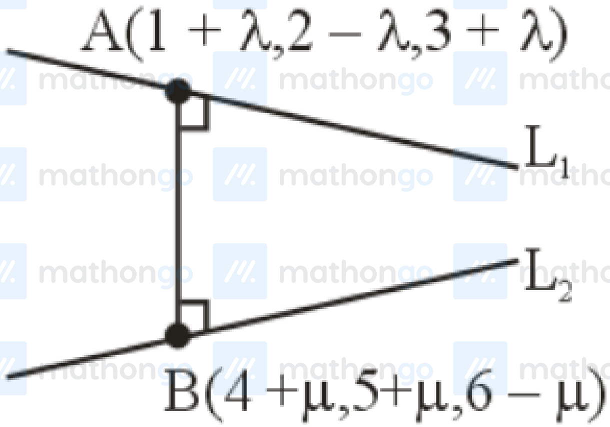
Passing points of lines L_1& L_2 are $(\lambda, 2, 1)\&(\sqrt{3}, 1, 2)$

$$S.D = \frac{\begin{vmatrix} \sqrt{3}-\lambda & -1 & 1 \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix}}$$

$$1 = \left|\frac{\sqrt{3}-\lambda}{\sqrt{3}}\right|$$

$\lambda = 0, \lambda = 2\sqrt{3}$

Q2



$\vec{b} = \hat{i} - \hat{j} + \hat{k}$ (DR 's of L_1)

$\vec{d} = \hat{i} + \hat{j} - \hat{k}$ (DR 's of L_2)

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$= 0\hat{i} + 2\hat{j} + 2\hat{k}$ (DR's of Line perpendicular to L_1 and L_2)

DR of AB line

$= (0, 2, 2) = (3+\mu-\lambda, 3+\mu+\lambda, 3-\mu-\lambda)$

$$\frac{3+\mu-\lambda}{0} = \frac{3+\mu+\lambda}{2} = \frac{3-\mu-\lambda}{2}$$

Solving above equation we get $\mu=-\frac{3}{2}$ and $\lambda=\frac{3}{2}$

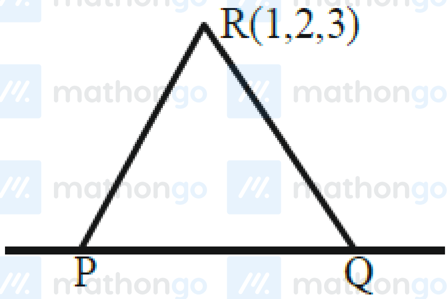
point $\mathrm{A}=\left(\frac{5}{2}, \frac{1}{2}, \frac{9}{2}\right)$

$\mathrm{B}=\left(\frac{5}{2}, \frac{7}{2}, \frac{15}{2}\right)$

Point of $\mathrm{AB}=\left(\frac{5}{2}, 2,6\right)=(\alpha, \beta, \gamma)$

$2(\alpha+\beta+\gamma)=5+4+12=21$

Q3



$\mathrm{P}(8 \lambda-3,2 \lambda+4,2 \lambda-1)$

$\mathrm{PR}=6$

$(8 \lambda-4)^2+(2 \lambda+2)^2+(2 \lambda-4)^2=36$

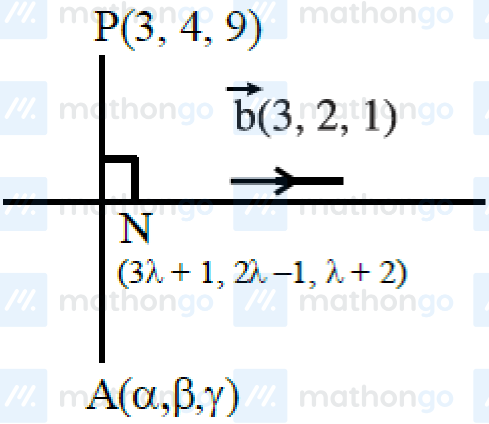
$\lambda=0,1$

Hence $\mathrm{P}(-3,4,-1) \& \mathrm{Q}(5,6,1)$

Centroid of $\triangle \mathrm{PQR}=(1,4,1) \equiv(\alpha, \beta, \gamma)$

$\alpha^2+\beta^2+\gamma^2=18$

Q4



$\overrightarrow{PN}\,\vec{b} = 0?$
$3(3\lambda - 2) + 2(2\lambda - 5) + (\lambda - 7) = 0$
$14\lambda = 23 \Rightarrow \lambda = \dfrac{23}{14}$
$\mathrm{N}\left(\dfrac{83}{14}, \dfrac{32}{14}, \dfrac{51}{14}\right)$
$\therefore \dfrac{\alpha + 3}{2} = \dfrac{83}{14} \Rightarrow \alpha = \dfrac{62}{7}$
$\frac{\beta+4}{2} = \frac{32}{14} \Rightarrow \beta = \frac{4}{7}$
$\frac{\gamma+9}{2} = \frac{51}{14} \Rightarrow \gamma = \frac{-12}{7}$
Ans. $14(\alpha + \beta + r) = 108$

Q5

$B = (2\lambda + 7, -3\lambda - 2, 6\lambda + 11)$

$(7, -2, 11)$

Point B lies on $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$
$\frac{2\lambda+7-6}{1} = \frac{-3\lambda-2-4}{0} = \frac{6\lambda+11-8}{3}$
$-3\lambda - 6 = 0$
$\lambda = -2$
$\mathrm{AB} = \sqrt{(7-3)^2 + (4+2)^2 + (11+1)^2}$
$= \sqrt{16 + 36 + 144}$
$= \sqrt{196} = 14$

Q6

$$\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$$

$$\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$$

the shortest distance between the lines

$$=\left|\frac{(\overrightarrow{\mathrm{a}}-\overrightarrow{\mathrm{b}})\cdot\left(\overrightarrow{\mathrm{d}_1}\times\overrightarrow{\mathrm{d}_2}\right)}{\left|\overrightarrow{\mathrm{d}_1}\times\overrightarrow{\mathrm{d}_2}\right|}\right|$$

$$=\left|\frac{\begin{vmatrix}\lambda-4 & 0 & 2\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}}{\left\|\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -3\\ 2 & 4 & -5\end{matrix}\right\|}\right|$$

$$=\left|\frac{(\lambda-4)(-10+12)-0+2(4-4)}{|2\hat{i}-1\hat{j}+0\hat{k}|}\right|$$

$$\frac{6}{\sqrt{5}}=\left|\frac{2(\lambda-4)}{\sqrt{5}}\right|$$

$$3=|\lambda-4|$$

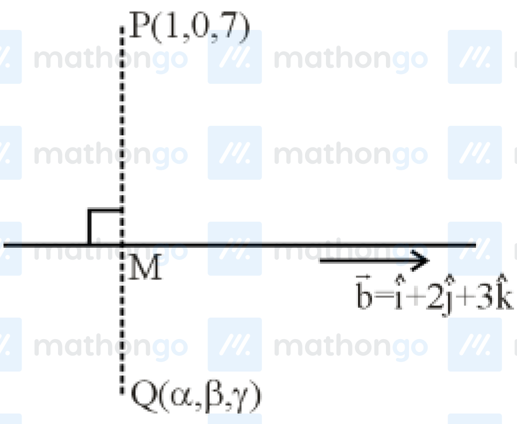
$$\lambda-4=\pm 3$$

$$\lambda=7,1$$

Sum of all possible values of λ is $=8$

Q7

$\mathrm{L}_1=\frac{\mathrm{x}}{1}=\frac{\mathrm{y}-1}{2}=\frac{\mathrm{z}-2}{3}=\lambda$



$M(\lambda, 1+2\lambda, 2+3\lambda)$

$\overrightarrow{PM} = (\lambda-1)\hat{i} + (1+2\lambda)\hat{j} + (3\lambda-5)\hat{k}$

$\overrightarrow{PM}$ is perpendicular to line L_1

$\overrightarrow{PM}\,\vec{b} = 0 \quad (\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k})$

$\Rightarrow \lambda - 1 + 4\lambda + 2 + 9\lambda - 15 = 0$

$14\lambda = 14 \Rightarrow \lambda = 1$

$\therefore M = (1, 3, 5)$

$\vec{Q} = 2\vec{M} - \vec{P}$ [M is midpoint of $\vec{P}$ & $\vec{Q}$]

$\vec{Q} = 2\hat{i} + 6\hat{j} + 10\hat{k} - \hat{i} - 7\hat{k}$

$\vec{Q} = \hat{i} + 6\hat{j} + 3\hat{k}$

$\therefore (\alpha, \beta, \gamma) = (1, 6, 3)$

Required line having direction cosine$(l,\ m, n)$

$l^2 + m^2 + n^2 = 1$

$$\Rightarrow l^2 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 = 1$$

$l^2 = \frac{1}{4}$

$\therefore l = \frac{1}{2}$ [Line make acute angle with x-axis]

Equation of line passing through $(1, 6, 3)$ will be

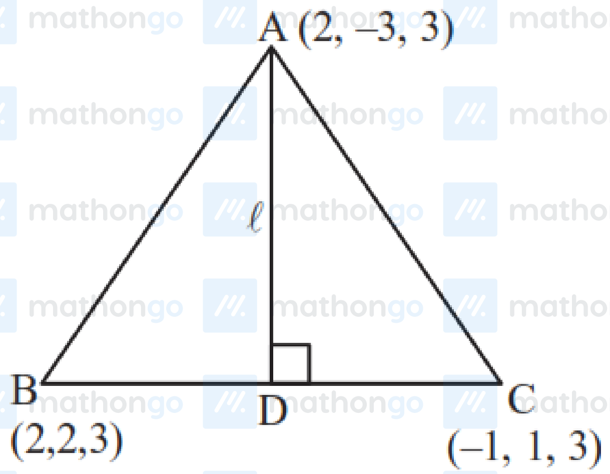
$\vec{r} = (\hat{i} + 6\hat{j} + 3\hat{k}) + \mu\left(\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}\right)$

Option 3 satisfying for $\mu = 4$

Q8

$AB = 5$

$AC = 5$



$\therefore$ D is midpoint of BC

$D\left(\frac{1}{2}, \frac{3}{2}, 3\right)$

$\therefore l=\sqrt{\left(2-\frac{1}{2}\right)^2+\left(-3-\frac{3}{2}\right)^2+(3-3)^2}$

$l=\sqrt{\frac{45}{2}}$

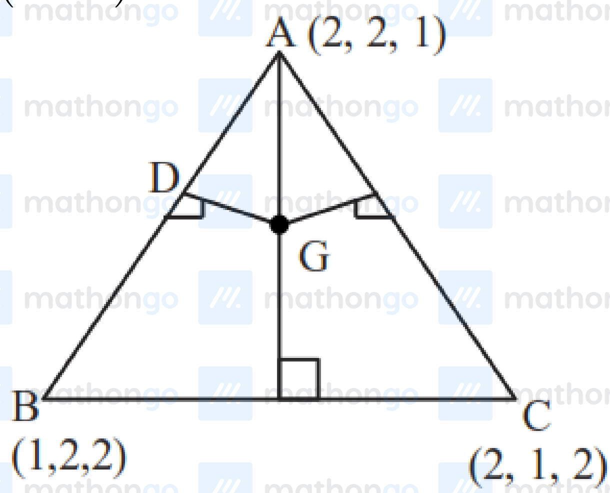
$\therefore 2l^2=45$

Q9

$\triangle$ABC is equilateral

Orthocentre and centroid will be same

$G\left(\frac{5}{3}, \frac{5}{3}, \frac{5}{3}\right)$



Mid-point of AB is D $\left(\frac{3}{2}, 2, \frac{3}{2}\right)$

$\therefore \ell_1 = \sqrt{\frac{1}{36} + \frac{1}{9} + \frac{1}{36}}$

$\ell_1 = \sqrt{\frac{1}{6}} = \ell_2 = \ell_3$

$\therefore \ell_1^2 + \ell_2^2 + \ell_3^2 = \frac{1}{2}$

Q10

Sol. $\frac{x-2}{1} = \frac{y}{-1} = \frac{z-7}{8} = \lambda$

$\frac{x+3}{4} = \frac{y+2}{3} = \frac{z+2}{1} = k$

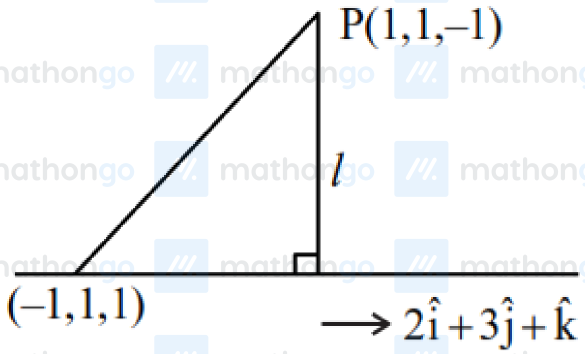
$\Rightarrow \lambda + 2 = 4k - 3$

$-\lambda = 3k - 2$

$\Rightarrow k = 1, \lambda = -1$

$8\lambda + 7 = k - 2$

$\therefore P = (1, 1, -1)$



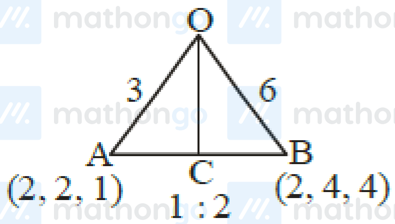
Projection of $2\hat{i} - 2\hat{k}$ on $2\hat{i} + 3\hat{j} + \hat{k}$ is

$= \frac{4-2}{\sqrt{4+9+1}} = \frac{2}{\sqrt{14}}$

$\therefore l^2 = 8 - \frac{4}{14} = \frac{108}{14}$

$\Rightarrow 14l^2 = 108$

Q11



length of $OC = \frac{\sqrt{136}}{3} = \frac{2\sqrt{34}}{3}$

Q12

Centroid G divides MR in 1 : 2

G(1, 2, 2)

Point of intersection A of given lines is (2,–6, 0)

$AG = \sqrt{69}$

Q13

Let $P(t, t-2, t)$ and $Q(2s-2, s, s)$

D.R's of PQ are 2, 1, 2

$$\frac{2s-2-t}{2} = \frac{s-t+2}{1} = \frac{s-t}{2}$$

$\Rightarrow t = 6$ and $s = 2$

$\Rightarrow P(6, 4, 6)$ and $Q(2, 2, 2)$

$$PQ : \frac{x-2}{2} = \frac{y-2}{1} = \frac{z-2}{2} = \lambda$$

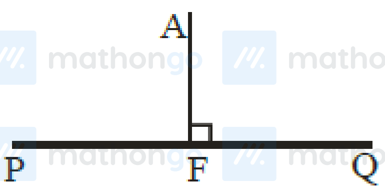
Let $F(2\lambda+2, \lambda+2, 2\lambda+2)$

$A(1, 2, 12)$

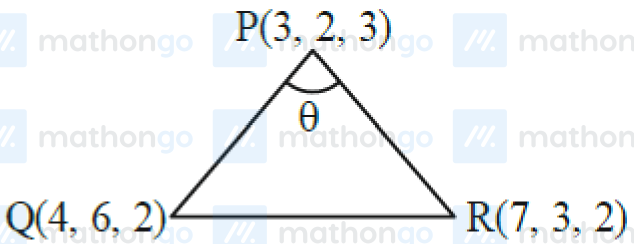
$\overrightarrow{AF} \cdot \overrightarrow{PQ} = 0$

$\therefore \lambda = 2$

So $F(6, 4, 6)$ and $AF = \sqrt{65}$



Q14



Direction ratio of PR $= (4, 1, -1)$

Direction ratio of PQ $= (1, 4, -1)$

Now, $\cos\theta = \left|\frac{4+4+1}{\sqrt{18}\cdot\sqrt{18}}\right|$

$\theta = \frac{\pi}{3}$

Q15

$L_1 : \dfrac{x-5}{4} = \dfrac{y-4}{1} = \dfrac{z-5}{3} = \lambda \quad \text{drs}(4,1,3) = b_1$

$M(4\lambda + 5, \lambda + 4, 3\lambda + 5)$

$L_2 : \dfrac{x+8}{12} = \dfrac{y+2}{5} = \dfrac{z+11}{9} = \mu$

$N(12\mu - 8, 5\mu - 2, 9\mu - 11)$

$\overrightarrow{MN} = (4\lambda - 12\mu + 13, \lambda - 5\mu + 6, 3\lambda - 9\mu + 16) \ldots (1)$

Now

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & 3 \\ 12 & 5 & 9 \end{vmatrix} = -6\hat{i} + 8\hat{k} \ldots (2)$$

Equation (1) and (2)

$\therefore \frac{4\lambda - 12\mu + 13}{-6} = \frac{\lambda - 5\mu + 6}{0} = \frac{3\lambda - 9\mu + 16}{8}$

I and II

$\lambda - 5\mu + 6 = 0 \ldots (3)$

I and III

$\lambda - 3\mu + 4 = 0 \ldots (4)$

Solve (3) and (4) we get

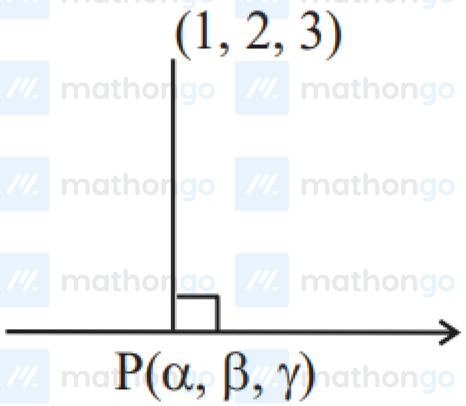
$\lambda=-1, \mu=1$

$\therefore \mathrm{M}(1,3,2)$

$\mathrm{N}(4,3,-2)$

$\therefore \overrightarrow{\mathrm{OM}} \cdot \overrightarrow{\mathrm{ON}}=4+9-4=9$

Q16



Let foot $P(5k-3,2k+1,3k-4)$

$\mathrm{DR's} \rightarrow \mathrm{AP}: 5\mathrm{k}-4, 2\mathrm{k}-1, 3\mathrm{k}-7$

DR's $\rightarrow$ Line: $5,2,3$

Condition of perpendicular lines $(25\mathrm{k}-20)+(4\mathrm{k}-2)+(9\mathrm{k}-21)=0$

Then $\mathrm{k}=\frac{43}{38}$

Then $19(\alpha+\beta+\gamma)=\mathbf{101}$

Q17

$\mathrm{L}_1: \frac{\mathrm{x}+1}{1}=\frac{\mathrm{y}}{1/2}=\frac{\mathrm{z}}{-1/12}, \mathrm{L}_2: \frac{\mathrm{x}}{1}=\frac{\mathrm{y}+2}{1}=\frac{\mathrm{z}-1}{\frac{1}{6}}$

$\mathrm{d}_1=$ shortest distance between $\mathrm{L}_1 \& \mathrm{L}_2$

$$=\left|\frac{(\vec{a}_2-\vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{\left|(\vec{b}_1 \times \vec{b}_2)\right|}\right|$$

$d_1=2$

$\mathrm{L}_3: \frac{\mathrm{x}-1}{2}=\frac{\mathrm{y}+8}{-7}=\frac{\mathrm{z}-4}{5}, \mathrm{L}_4: \frac{\mathrm{x}-1}{2}=\frac{\mathrm{y}-2}{1}=\frac{\mathrm{z}-6}{-3}$

$d_2 =$ shortest distance between L_3 & L_4

$d_2 = \frac{12}{\sqrt{3}}$ Hence

$= \frac{32\sqrt{3}\, d_1}{d_2} = \frac{32\sqrt{3}\times 2}{\frac{12}{\sqrt{3}}} = 16$

Q18

$L_1 \perp L_2$

$3 - 1 + 2P = 0$

$P = -1$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix} = -\hat{i} + 7\hat{j} + 4\hat{k}$$

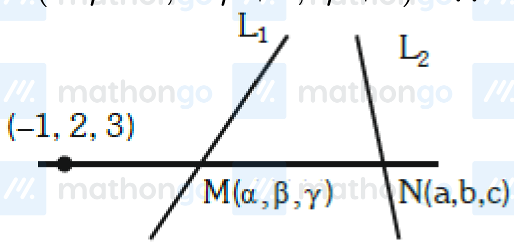
$\therefore (-\delta, 7\delta, 4\delta)$ will lie on L_3

For $\delta = 1$ the point will be $(-1, 7, 4)$

Q19

$M(3\lambda + 1, 2\lambda + 2, -2\lambda - 1) \quad \therefore \alpha + \beta + \gamma = 3\lambda + 2$

$N(-3\mu - 2, -2\mu + 2, 4\mu + 1) \quad \therefore a + b + c = -\mu + 1$



$$\frac{3\lambda + 2}{-3\mu - 1} = \frac{2\lambda}{-2\mu} = \frac{-2\lambda - 4}{4\mu - 2}$$

$3\lambda\mu + 2\mu = 3\lambda\mu + \lambda$

$2\mu = \lambda$

$2\lambda\mu - \lambda = \lambda\mu + 2\mu$

$\lambda\mu = \lambda + 2\mu$

$\Rightarrow \lambda\mu = 2\lambda$

$\Rightarrow \mu = 2 \quad (\lambda \neq 0)$

$\therefore \quad \lambda = 4$

$\alpha + \beta + \gamma = 14$

$a + b + c = -1$

$$\frac{(\alpha + \beta + \gamma)^2}{(a + b + c)^2} = 196$$

Q20

A vector in the direction of the required line can be obtained by cross product of

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ -1 & 3 & 2 \end{vmatrix}$$

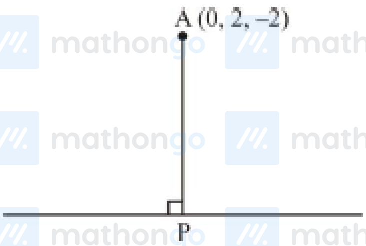
$= -9\hat{i} - 9\hat{j} + 9\hat{k}$

Required line,

$\overrightarrow{\mathrm{r}} = (5\hat{\mathrm{i}} - 4\hat{\mathrm{j}} + 3\hat{\mathrm{k}}) + \lambda'(-9\hat{\mathrm{i}} - 9\hat{\mathrm{j}} + 9\hat{\mathrm{k}})$

$\overrightarrow{\mathrm{r}} = (5\hat{\mathrm{i}} - 4\hat{\mathrm{j}} + 3\hat{\mathrm{k}}) + \lambda(\hat{\mathrm{i}} + \hat{\mathrm{j}} - \hat{\mathrm{k}})$

Now distance of $(0, 2, -2)$



P.V. of P $\equiv (5+\lambda)\hat{i} + (\lambda - 4)\hat{j} + (3-\lambda)\hat{k}$

$\overrightarrow{\mathrm{AP}} = (5+\lambda)\hat{i} + (\lambda - 6)\hat{j} + (5-\lambda)\hat{k}$

$\overrightarrow{\mathrm{AP}} \cdot (\hat{i} + \hat{j} - \hat{k}) = 0$

$5 + \lambda + \lambda - 6 - 5 + \lambda = 0$

$\lambda = 2$

$|\overrightarrow{\mathrm{AP}}| = \sqrt{49 + 16 + 9}$

$|\overrightarrow{\mathrm{AP}} = \sqrt{74}$

Q21

$\frac{x}{1}=\frac{y}{1}=\frac{z-1}{0}=r \rightarrow Q(r, r, 1)$

$\frac{x}{1}=\frac{y}{-1}=\frac{z+1}{0}=k \rightarrow R(k,-k,-1)$

$\overline{PQ}=(a-r) \hat{i}+(a-r) \hat{j}+(a-1) \hat{k}$

$a=r+a-r=0$

$2a=2r \rightarrow a=r$

$\overline{PR}=(a-k) i+(a+k) \hat{j}+(a+1) \hat{k}$

$a-k-a-k=0 \Rightarrow k=0$

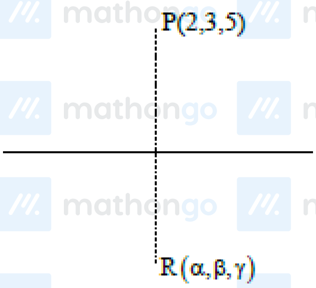
$As, PQ \perp PR$

$(a-r)(a-k)+(a-r)(a+k)+(a-1)(a+1)=0$

$a=1$ or -1

$12 a^2=12$

Q22



$\because \overrightarrow{\mathrm{PR}} \perp(2,3,4)$

$\therefore \overrightarrow{\mathrm{PR}} \cdot(2,3,4)=0$

$(\alpha-2, \beta-3, \gamma-5) \cdot(2,3,4)=0$

$\Rightarrow 2 \alpha+3 \beta+4 \gamma=4+9+20=33$

Q23

$$L_2 = \frac{x+4}{3} = \frac{y-4}{2} = \frac{z-3}{0}$$

$$\therefore \mathrm{S.D} = \frac{\begin{vmatrix} \mathrm{x}_2 - \mathrm{x}_1 & \mathrm{y}_2 - \mathrm{y}_1 & \mathrm{z}_2 - \mathrm{z}_1 \\ 2 & -3 & 2 \\ 3 & 2 & 0 \end{vmatrix}}{\left|\overrightarrow{\mathrm{n}_1} \times \overrightarrow{\mathrm{n}_2}\right|}$$

$$= \frac{\begin{vmatrix} 5 & -5 & -7 \\ 2 & -3 & 2 \\ 3 & 2 & 0 \end{vmatrix}}{\left|\overrightarrow{\mathrm{n}_1 \times \mathrm{n}_2}\right|}$$

$$= \frac{141}{-4\hat{i} + 6\hat{j} + 13\hat{k}}$$

$$= \frac{141}{\sqrt{16+36+169}}$$

$$= \frac{141}{\sqrt{221}}$$

Q24

$$\frac{x-4}{12} = \frac{x+6}{4} = \frac{z+2}{6}$$

$$\frac{x-4}{\frac{6}{7}} = \frac{y+6}{\frac{2}{7}} = \frac{z+2}{\frac{3}{7}} = 21$$

$$\left(21 \times \frac{6}{7} + 4, \frac{2}{7} \times 21 - 6, \frac{3}{7} \times 21 - 2\right)$$

$$= (22, 0, 7) = (a, b, c)$$

$$\therefore \sqrt{324 + 144 + 16} = 22$$